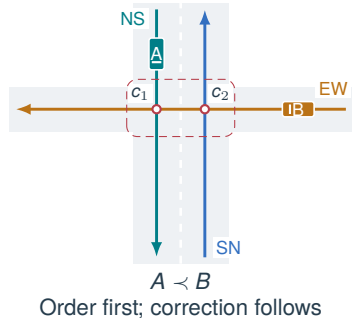


Decision-Barrier Framework

Barrier-Based Platoon Control for
Autonomous Intersection Management

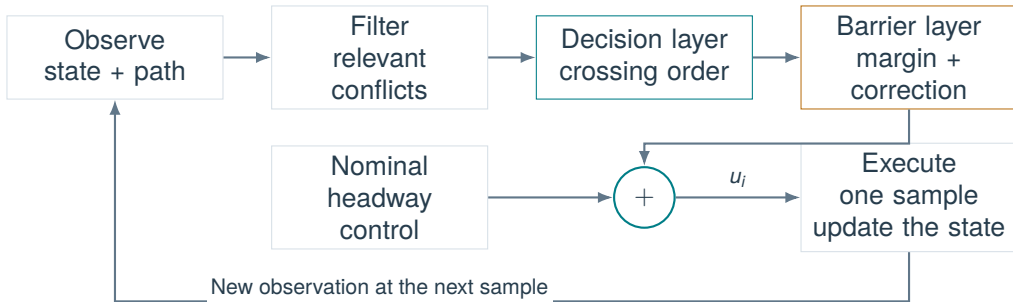
How are crossing order and speed
correction connected?

Method explanation and proposed experiments



Two layers, one feedback loop

The decision layer chooses who yields



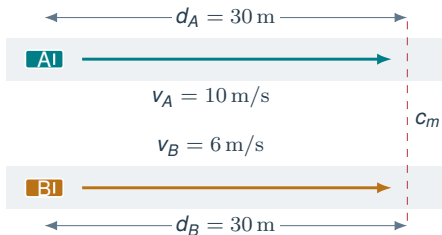
Spatial filter

Inside the approach window and sharing a conflict.

Time-margin test

Apply the activation threshold only to these pairs.

Arrival time adds the missing speed information



Same distance, different predicted arrivals

$$T_i^m = \frac{d_i^m}{\max(v_i, v_{\min})}$$

$$T_A = \frac{30}{10} = 3 \text{ s},$$

$$T_B = \frac{30}{6} = 5 \text{ s}.$$

Distance follows the planned path.
Prediction assumes the current speed.

Arrival-time estimate is not collision TTC. This example uses speeds above the floor.

The decision layer selects a crossing order

Start with arrival time

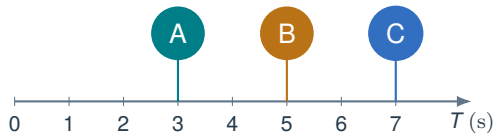
$$J_i^m = T_i^m, \quad \mathcal{O}^m = (A, B, C)$$

A smaller score gives higher priority.

Extensions in the draft

$$J_i^m = \lambda_1 T_i^m + \lambda_2 Q_{l_i} + \lambda_3 P_{l_i} + \lambda_4 R_i$$

Q : queue length; P : road priority;
 R : maneuver priority.



$$A \prec B \prec C$$

The barrier layer acts on
the selected order

Tie-breaking, order retention, and consistency across conflict points still require explicit implementation rules.

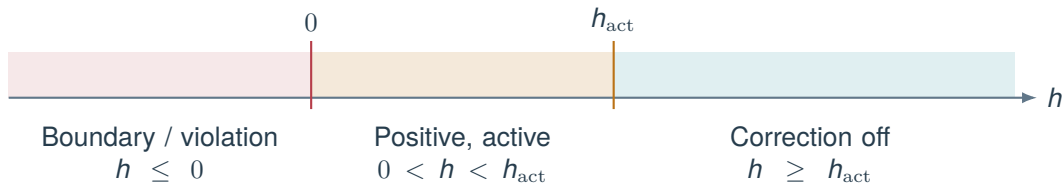
Safety margin and activation threshold

Symmetric separation

$$h_{ij}^{m,\text{sym}} = |T_i^m - T_j^m| - T_{\text{safe}}$$

After choosing $a \prec b$

$$h_{ab}^{m,\text{ord}} = T_b^m - T_a^m - T_{\text{safe}}$$



Activation starts before the margin reaches zero

Apply this test only to conflict-relevant vehicles. Ordered and symmetric margins are distinct definitions.

Smaller positive margins produce larger corrections

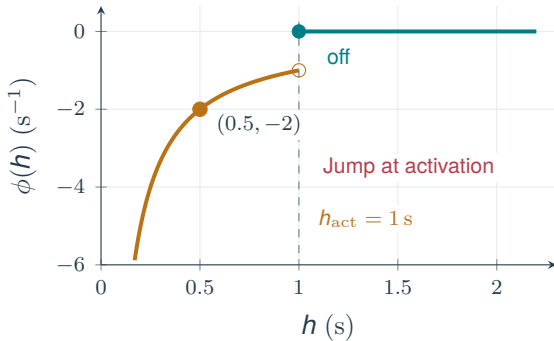
Original heuristic expression

$$\phi(h) = \begin{cases} -1/h, & h < h_{\text{act}}, \\ 0, & h \geq h_{\text{act}}. \end{cases}$$

On $0 < h < h_{\text{act}}$, a smaller margin gives a stronger negative correction.

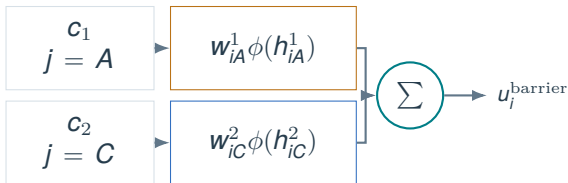
Undefined at $h = 0$; sign reversal for $h < 0$.

The plotted domain is $h > 0$. Threshold is illustrative.



From pairwise terms to vehicle control

$$u_i = \underbrace{u_i^{\text{headway}}}_{\text{nominal control}} + \underbrace{\sum_{m \in \mathcal{C}_i} \sum_{j \in \mathcal{N}_i^m} w_{ij}^m \phi(h_{ij}^m)}_{u_i^{\text{barrier}}}$$



$\mathcal{C}_i$: relevant conflict points.

$\mathcal{N}_i^m$: conflicting vehicles at c_m .

The order determines who yields. In the example, the first vehicle has zero correction weight.

For acceleration input: $[u] = \text{m/s}^2$, $[\phi] = \text{s}^{-1}$; w includes an m/s gain. The sum is the draft's symmetric-margin form; order-aware use must retain the selected precedence.

Worked example: prediction and correction

1. Observe and predict

	$d(\text{m})$	$v(\text{m/s})$	$T(\text{s})$
A	30	10	3
B	50	10	5

$$A \prec B$$

$$T_{\text{safe}} = 1.5 \text{ s}, \quad h_{\text{act}} = 1 \text{ s}$$

Illustrative parameters; not calibrated settings.

2. Evaluate the margin

$$h = 5 - 3 - 1.5 = 0.5 \text{ s} < h_{\text{act}}$$

3. Apply correction to the later vehicle

$$\phi = -\frac{1}{0.5} = -2 \text{ s}^{-1}$$

Assume $u^{\text{headway}} = 0$, $w_A = 0$, $w_B = 0.5 \text{ m/s}$.

$$u_A = 0, \quad u_B = -1 \text{ m/s}^2$$

One feedback step increases the predicted margin

1 Predict

$$T_A = 3 \text{ s}$$

$$T_B = 5 \text{ s}$$

$$h = 0.50 \text{ s}$$



2 Brake B

$$\Delta t = 0.1 \text{ s}$$

$$u_A = 0$$

$$u_B = -1 \text{ m/s}^2$$



3 Recompute

$$T_A^+ = 2.9 \text{ s}$$

$$T_B^+ = 4.95 \text{ s}$$

$$h^+ = 0.55 \text{ s}$$

$$d^+ = d - v\Delta t - \frac{1}{2}u\Delta t^2, \quad v^+ = v + u\Delta t$$

	d^+ (m)	v^+ (m/s)	T^+ (s)
A	29	10	2.9
B	49.005	9.9	4.95

$$h^+ = 4.95 - 2.9 - 1.5$$

Constant acceleration within this sample. The correction remains active; one improved prediction is not a safety proof.

Boundaries of the original expression

Formula and actuation

- $h = 0$: undefined. $h < 0$: sign reversal.
- A speed floor prevents division by zero, but does not validate stopped-vehicle predictions.
- Input limits and switching can alter the intended response.

Geometry and candidate selection

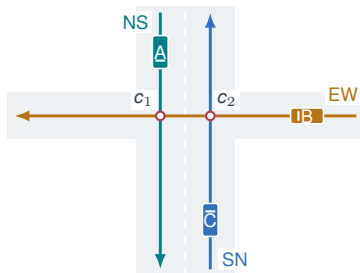


Point arrival times omit finite occupancy.
Keeping only lane leaders also needs verification.

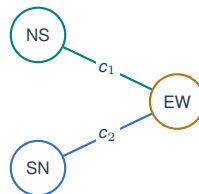
Later direction: a CBF-QP safety filter

The draft proposes optimization-based filtering. Feasibility, input bounds, and model assumptions must be analyzed before safety claims.

Case A: three directed lanes, two crossing conflicts



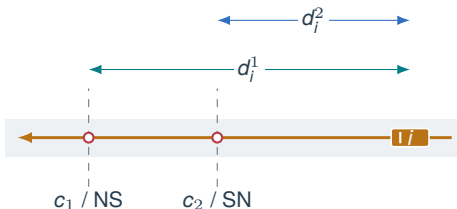
Bidirectional NS/SN; unidirectional EW.



	NS	SN	EW
NS	0	0	1
SN	0	0	1
EW	1	1	0

Separated, straight lanes. NS and SN have no crossing-conflict edge; same-lane following is handled by the nominal layer.

Case A: the EW vehicle checks both points



Use a separate path distance and arrival prediction for each point.

Pairwise correction, then aggregation

$$u_{\text{EW}}^{\text{barrier}} = u_{\text{EW,NS}}^{\text{barrier}} + u_{\text{EW,SN}}^{\text{barrier}}$$

Each term includes activation and the selected yielding role.

Priorities at the two points must be consistent; the draft leaves this coordination rule open.

Signed braking example $(-1, -3) \text{ m/s}^2$

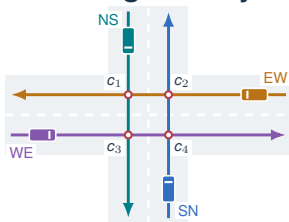
$$\underbrace{-1 + (-3)}_{\text{sum}} = -4$$

$$\underbrace{\min(-1, -3)}_{\text{strongest individual braking}} = -3$$

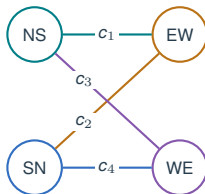
$\max(-1, -3) = -1$ gives weaker braking. Aggregation alone does not enforce actuator limits or prove safety.

Case B: four directed lanes and a conflict matrix

Road geometry



Conflict graph



Matrix A

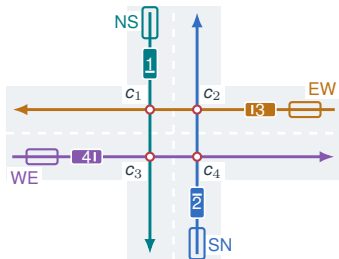
$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Row/column order:
NS, SN, EW, WE

Four unique edges; each appears twice in the symmetric matrix

c_1 : NS–EW; c_2 : SN–EW; c_3 : NS–WE; c_4 : SN–WE. Turning movements require a new conflict map.

Case B: candidate selection and online calculation

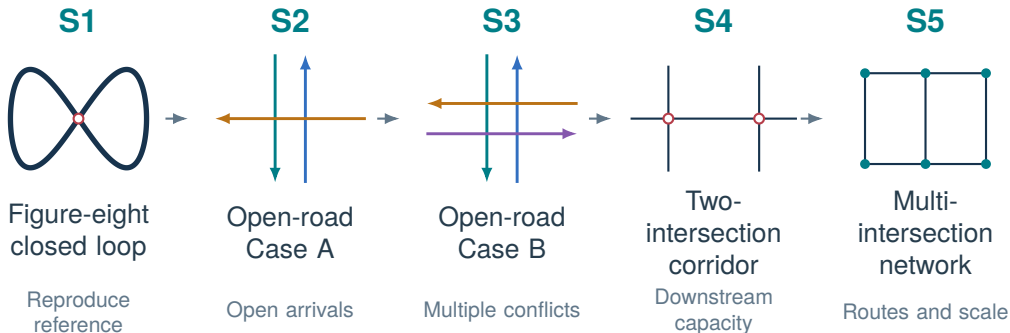


Filled: leader candidate. Outline: following vehicle.

1. Select the nearest approaching vehicle on each nonempty lane.
2. Use A to select $(1, 3), (2, 3), (1, 4), (2, 4)$.
3. Predict arrivals, choose order, compute h , and test activation.
4. Aggregate per vehicle, execute, then update the candidates.

Implementation requirement beyond the leader-only rule: retain occupying vehicles until rear clearance; validate that candidate reduction omits no relevant conflicts.

Five experimental stages



Proposed test sequence. Each stage expands the interaction structure; no experimental outcome is assumed.

Open-road single-intersection tests

Matched arrivals for S2 and S3

Random arrivals



Burst / near-simultaneous



S2: three incoming directions.

S3: four incoming directions.

Use the same arrival and route files for each controller.

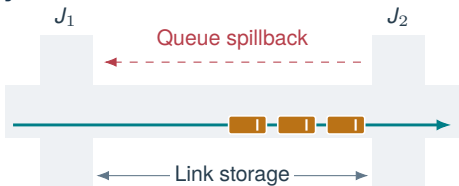
Use paired random seeds and report variability. Count requested, admitted, and completed vehicles, including uninserted and unfinished demand.

Sweep demand and stress the controller

- Balanced flows, then one dominant approach.
- Near ties, bursts, and low-speed queues.
- Observe margins, physical conflicts, delay, stops, and remaining queues.

Corridor and network tests

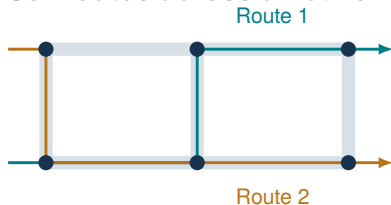
S4: finite storage between two junctions



Track platoon discharge, travel time, and downstream queues. Can local decisions respect downstream capacity?

Downstream admission and inter-junction coordination are future design tasks, not existing controller capabilities.

S5: routes across a network



Increase network size, demand, and active conflicts. Measure trip delay, queues, deadlock, and computation time.

Baselines and key performance indicators

Comparison methods

- Fully actuated signal control (FAS).
- Unsignalized priority control.
- Original distance-difference barrier, where reproducible.

Ablations

Weighted sum / strongest signed braking / CBF-QP. Keep the same crossing-order rule where feasible.

Measures and units

Efficiency	Throughput (veh/h); mean / total delay (s/veh, s)
Energy	Energy per vehicle (Wh/veh), with a specified model
Comfort	Acceleration / jerk RMS (m/s^2 , m/s^3); stops/veh
Safety	Collisions, occupancy conflicts; minimum margin (s)
Robustness	Deadlock count; computation time (ms)

Match geometry, demand, vehicle limits, and seeds. Arrival-time estimate is not collision TTC. Outcomes remain to be measured.

Sources: draft, Experiments; [SUMO: traffic lights](#); [TripInfo](#); [Ames et al. \(2017\)](#), CBF-QP.